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(54) **TOWABLE ASSET WITH AUTOMATED MONITORING**

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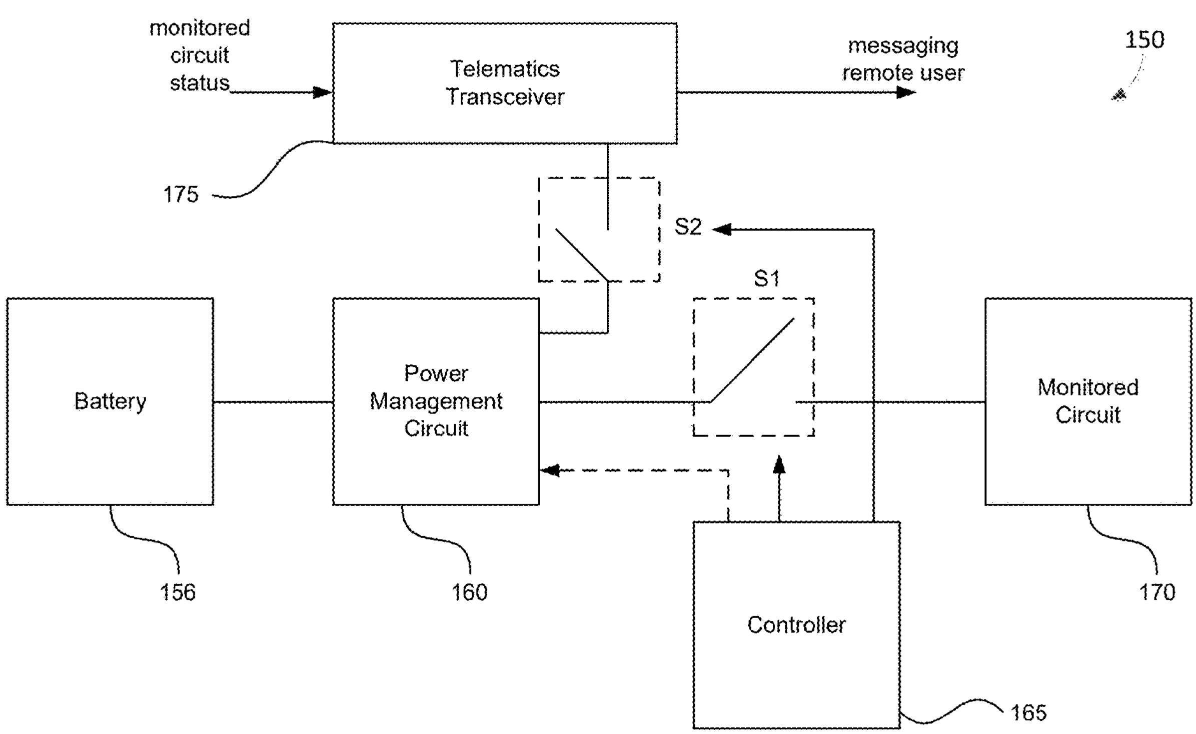
(57) **ABSTRACT**

An automated asset monitoring system is provided for a towable asset. The system includes a power supply voltage that may be converted to power a monitored circuit during an active mode of operation while the towable asset is not connected to a tractor. The system powers the monitored circuit during the active mode so that a message regarding an operating condition of the monitored circuit may be transmitted by a telematics transceiver to a remote user.



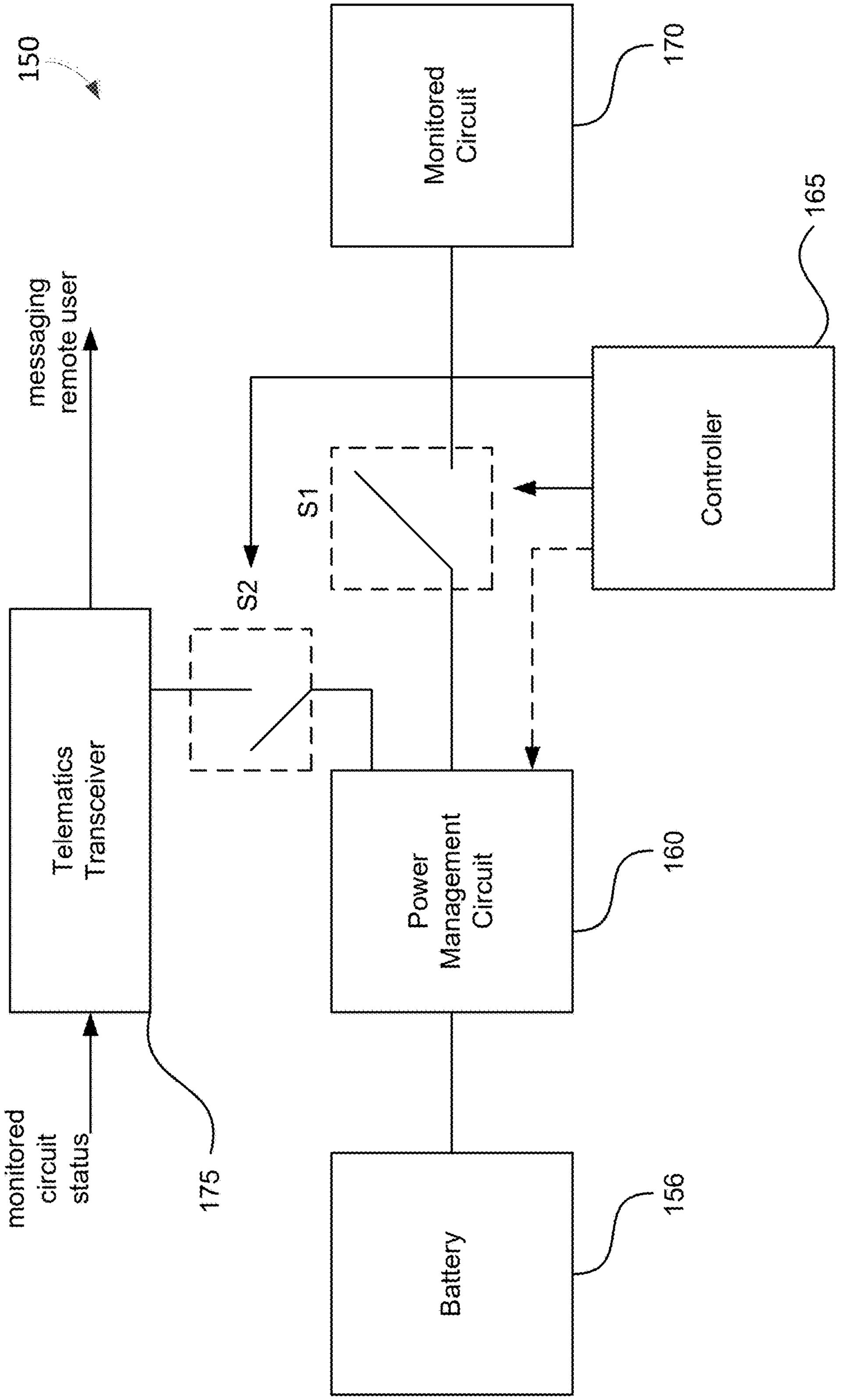

150
messaging remote user
Monitored Circuit
170
165
Controller
S1
S2
Telematics Transceiver
Power Management Circuit
160
monitored circuit status
175
Battery
156

FIG. 1A

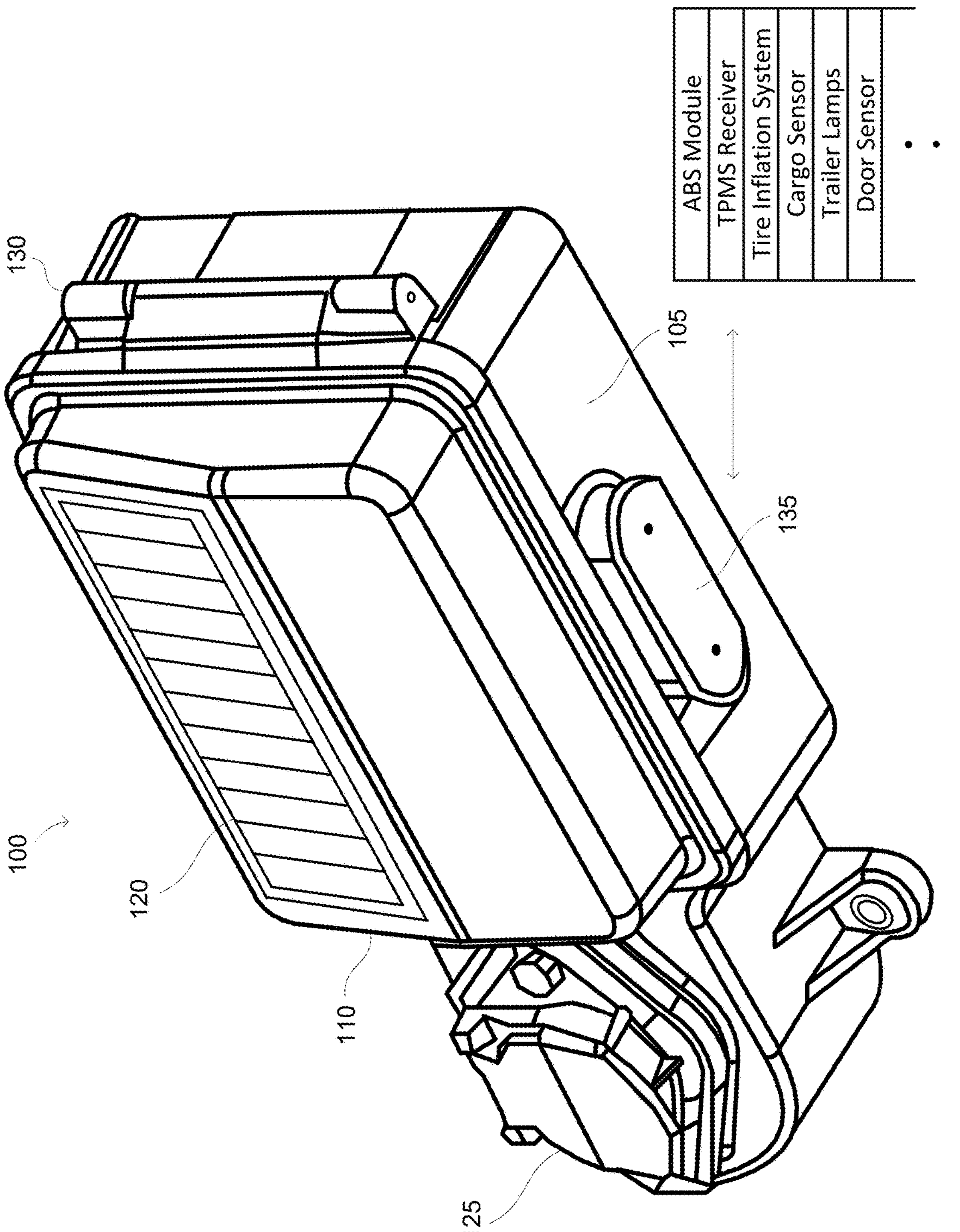

100
105
110
120
125
130
135
ABS Module
TPMS Receiver
Tire Inflation System
Cargo Sensor
Trailer Lamps
Door Sensor
. . .

FIG. 1B

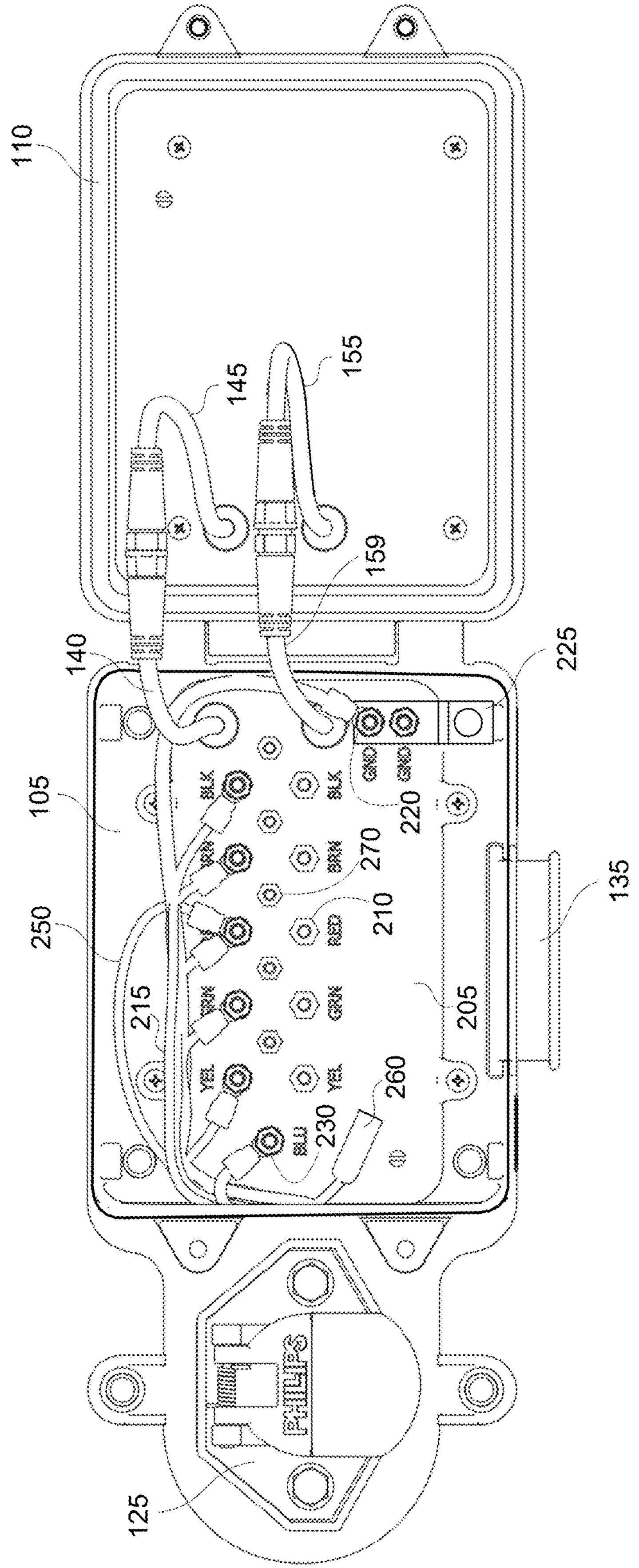

110
145
155
159
140
225
105
220
270
210
250
215
205
135
260
230
125
BLK
BRN
RED
GRN
YEL
BLU
GND
PHILIPS

FIG. 2

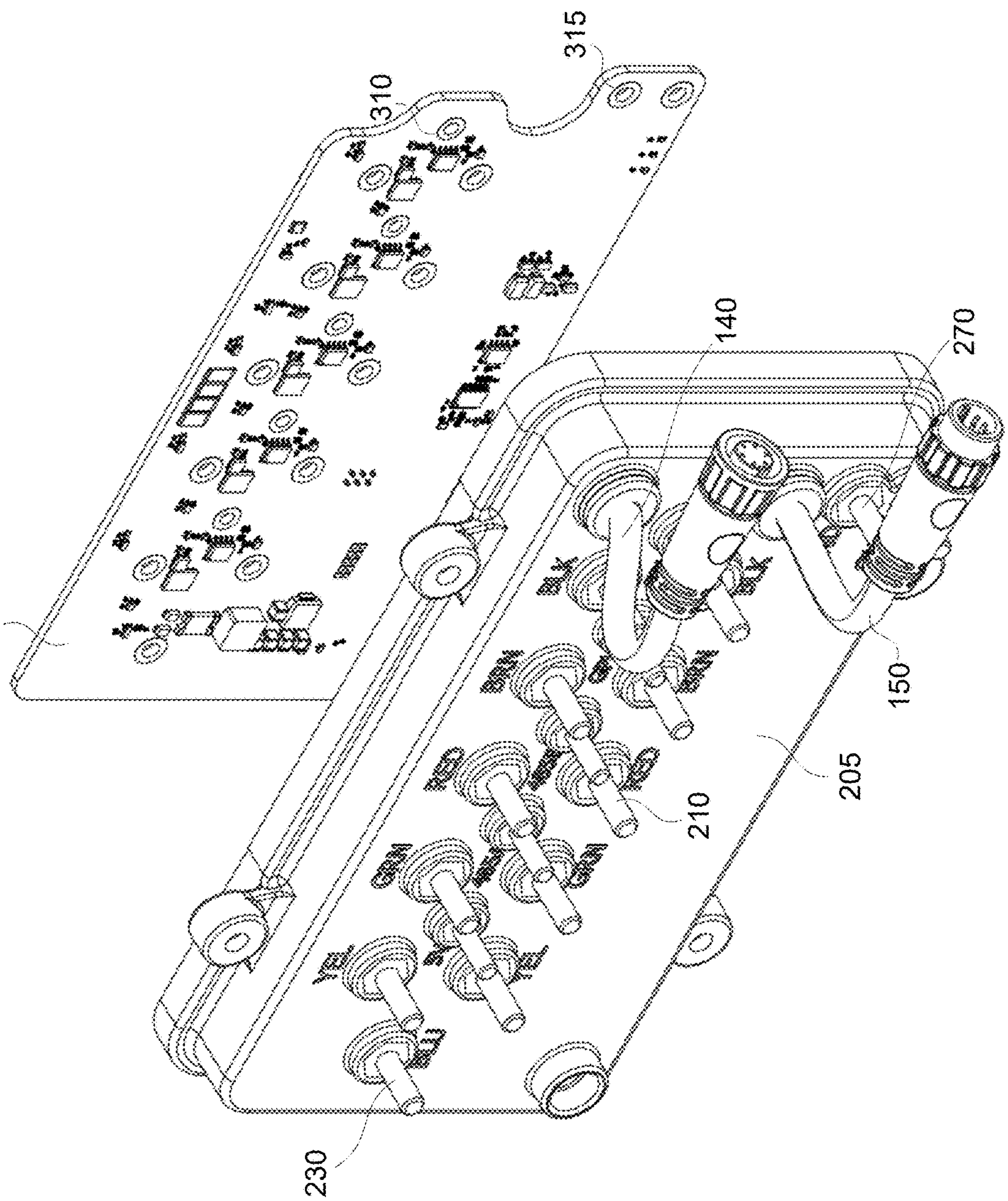


FIG. 3A

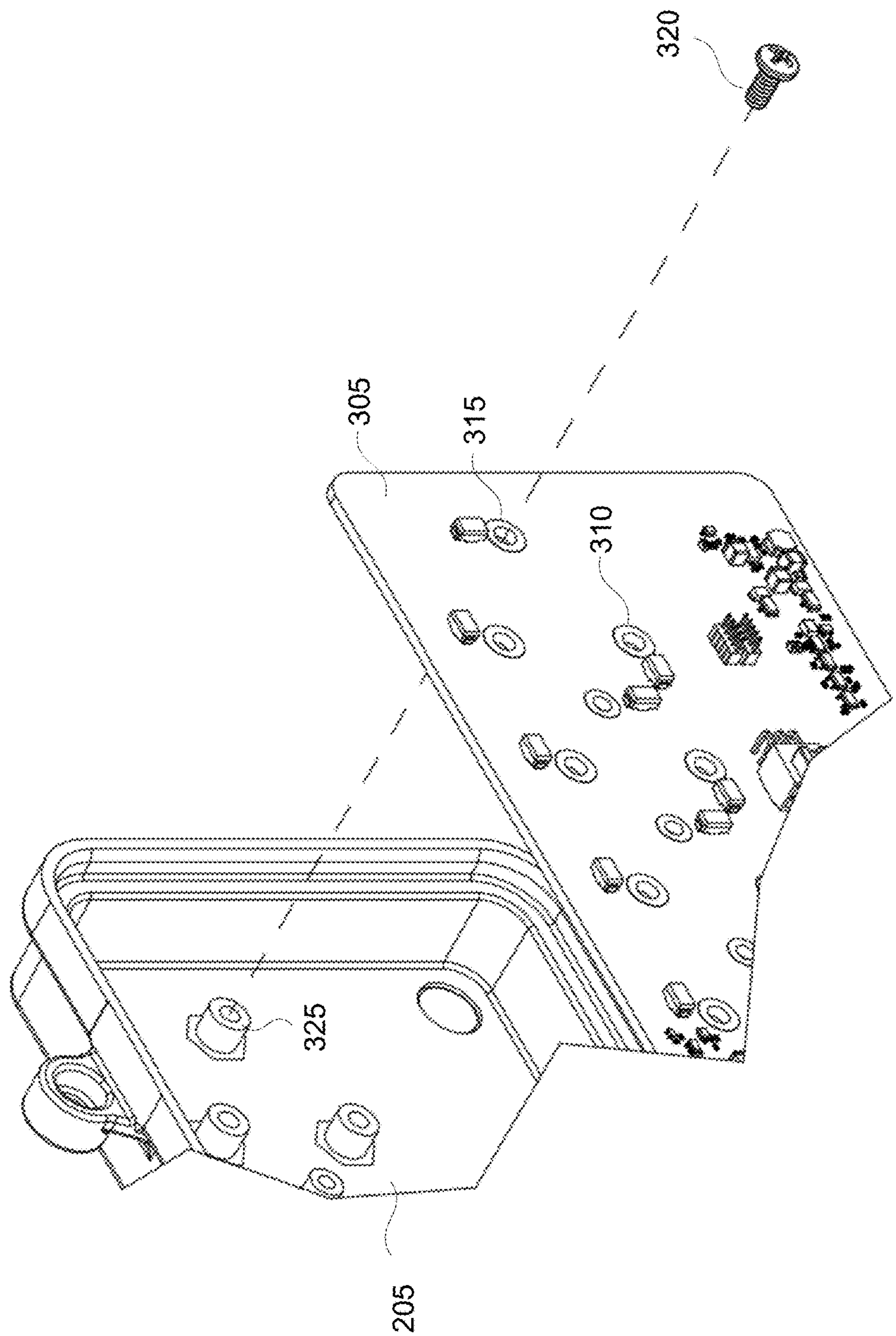


FIG. 3B

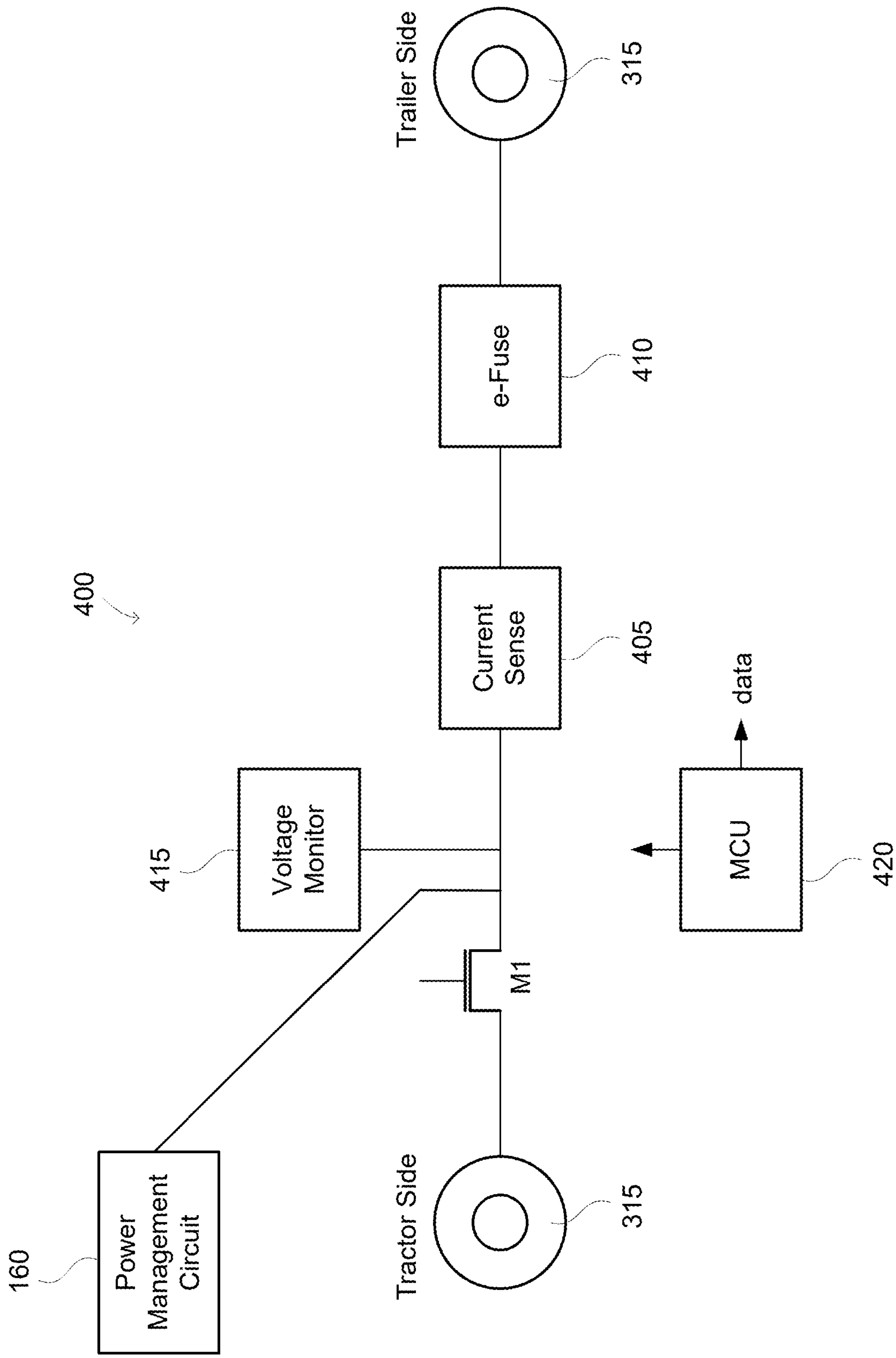

400
Trailer Side
315
e-Fuse
410
Current Sense
405
Voltage Monitor
415
MCU
420
data
M1
Tractor Side
315
Power Management Circuit
160

FIG. 4

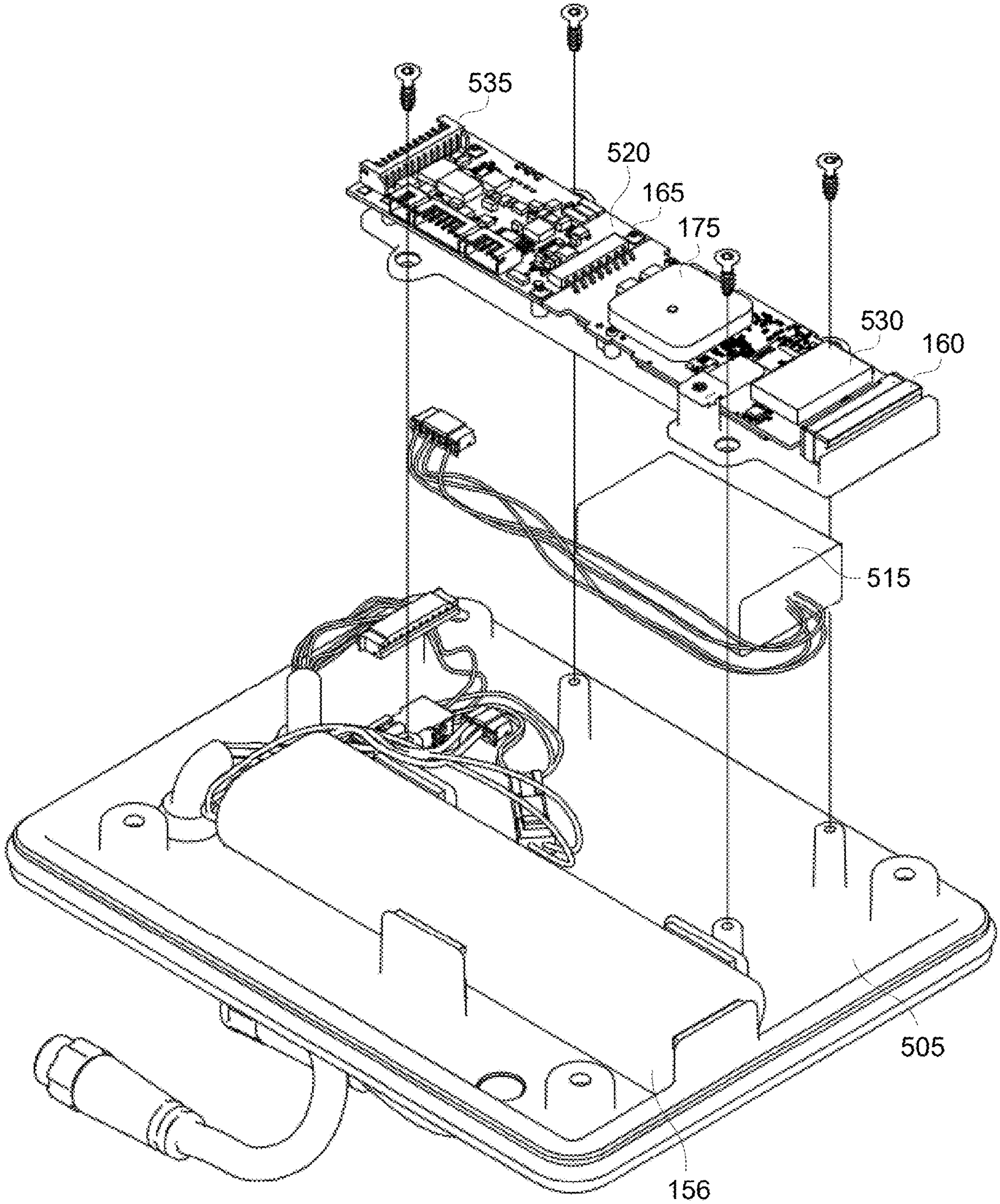


FIG. 5

TOWABLE ASSET WITH AUTOMATED MONITORING

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a Continuation of U.S. patent application Ser. No. 17/890,916, filed Aug. 18, 2022, which is a Continuation of International Application No. PCT/IB2021/051563, filed Feb. 25, 2021, which in turn claims the benefit of U.S. Provisional Application No. 62/980,011, filed on Feb. 21, 2020, each of which are hereby incorporated herein by reference in its entirety as if fully set forth below and for all applicable purposes.

TECHNICAL FIELD

[0002] The present invention relates to the monitoring of a towable asset, and more particularly, to an automated monitoring of a towable asset.

BACKGROUND

[0003] The design of the tractor in a tractor-trailer varies depending upon the region but regardless of the design, it is conventional to use a J-560 interface (commonly referred to as a 7-way coupler) on the semi-trailer or other type of towable asset to ease the connection and disconnection of the trailer from the tractor. As implied by the name, a 7-way coupler includes 7 pins or terminals. One pin is for ground. The remaining six pins are conventionally used to drive the tail and running lights and marker light, the license plate lamp, an auxiliary circuit such as an anti-lock brake (ABS) unit, the left turn signal, the right turn signal, and the stop lights, respectively.

[0004] The proper functioning of these various lights and circuits is vital for public safety. Various monitors have thus been developed that may wirelessly report the operating status of the lights and circuits in the trailer. But the ease of integration of such monitors is an issue. Similarly, various telematic reporting systems have been developed for the trucking industry. Customers may have thousands of rigs so the costs of retrofitting them to include monitors and telematics must be minimized.

SUMMARY

[0005] In accordance with an aspect of the disclosure, an automated towable asset monitoring system having an active mode of operation in which a towable asset is not coupled through a 7-way interface to a tractor and an inactive mode of operation in which the towable asset is coupled through the 7-way interface to a tractor is provided that includes a battery; a power management circuit configured to convert a battery voltage from the battery to a monitored circuit power supply voltage; a controller configured to command the monitored circuit power supply voltage to power a monitored circuit in the towable asset during the active mode of operation and to prevent the monitored circuit power supply voltage to power the monitored circuit during the inactive mode of operation; and a telematics transceiver configured to transmit data regarding an operation of the monitored circuit during the active mode of operation to a user remote from the towable asset.

[0006] In accordance with another aspect of the disclosure, a method of monitoring a monitored circuit in a towable asset while the towable asset is without an auxiliary power from a tractor, the method comprising: generating a monitored circuit power supply voltage from a battery; powering the monitored circuit using the monitored circuit power supply voltage to cause the monitored circuit to operate; and transmitting data regarding an operating condition of the monitored circuit from a telematics transceiver to a user.

[0007] In accordance with yet another aspect of the disclosure, a system for monitoring a lamp in a towable asset is provided that includes: a rechargeable battery; a power management circuit configured to charge the rechargeable battery using auxiliary power while the towable asset is connected to a tractor, the power management circuit being further configured to generate a lamp power supply voltage during an active mode of operation in which the towable asset does not receive the auxiliary power; a monitoring circuit configured to measure the lamp power supply voltage and a current delivered to the lamp to determine an operating condition of the lamp; a controller configured to command the power management circuit to supply the lamp power supply voltage to the monitoring circuit during the active mode of operation, and a telematics transceiver configured to transmit a message regarding the operating condition to a user remote from the towable asset.

[0008] These and additional advantageous features of the disclosed embodiments may be better appreciated through a consideration of the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1A is a high-level diagram of an automated asset monitoring system in accordance with an aspect of the disclosure.

[0010] FIG. 1B is a perspective view of a nose box with automated asset monitoring in accordance with an aspect of the disclosure.

[0011] FIG. 2 is a plan view of the nose box of FIG. 1B with its lid opened in accordance with an aspect of the disclosure.

[0012] FIG. 3A is a perspective exploded view of a post platform and the associated circuit board for a nose box with automated asset monitoring in accordance with an aspect of the disclosure.

[0013] FIG. 3B is a perspective view exploded view, partially cut-away, of the trailer side of the post platform and circuit board of FIG. 3A.

[0014] FIG. 4 illustrates a lamp monitoring circuit for the circuit board of FIGS. 3A and 3B in accordance with an aspect of the disclosure.

[0015] FIG. 5 is a trailer-side perspective exploded view of an inner cover for a nose box lid and associated circuitry in accordance with an aspect of the disclosure.

[0016] Embodiments of the present disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures.

DETAILED DESCRIPTION

[0017] To provide improved monitoring and telematics, an automated asset monitoring system is provided for the monitoring of a towable asset while the towable asset is not receiving the auxiliary power from a 7-way connection to the tractor in situations where the tractor is not connected to the towable asset. An example automated asset monitoring system **150** is shown in FIG. **1**A. Through system **150**, a user may remotely power various circuits within a towable asset despite the towable asset being disconnected from the tractor and thus without the auxiliary power provided by the tractor's J-560 coupler (commonly denoted as a 7-way coupler). A battery **156** functions as a power source for system **150** with regard to the selective powering and/or activation of a monitored circuit **170** in the towable asset. As used herein, the term "trailer" is used generically to refer to a semi-trailer, a chassis, a flat bed, or any other suitable towable asset that may be towed by the tractor and receive the tractor's 7-way coupler. In that regard, the terms "trailer" and "towable asset" are used interchangeably herein.

[0018] The monitored circuit **170** may be a lamp within the trailer such as one of the lamps discussed earlier that are powered by the 7-way connection to the tractor. These lamps are powered by a 12V power signal. Since battery **156** (e.g., a rechargeable battery such as a rechargeable lithium-ion battery) will typically provide a battery voltage that is lower than 12 V (e.g., 4 V), a power management circuit **160** may function as a DC-DC boost converter to boost the battery voltage to a 12 V power signal should the monitored circuit **170** be powered by a 12 V power supply. The following discussion will assume that each monitored circuit **170** is amenable to being powered by a 12 V power supply voltage as this is typical for trucking sensors. However, it will be appreciated that a monitored circuit **170** may require some other level of power supply voltage in alternative embodiments. In contrast to the monitored circuit **170**, a telematics transceiver **175** will typically be powered by a telematics supply voltage that is lower than the battery voltage. For such assets, power management circuit **160** may function as a DC-DC buck converter to reduce the battery voltage to a telematics supply voltage for the telematics transceiver **175** and related circuitry. If, on the other hand, the telematics transceiver **175** may be powered by 12 V, power management circuit **160** need only provide the 12 V power supply voltage and would have no need of generating a telematics power supply voltage. In alternative embodiments, power management circuit **160** may generate more than two different power supply voltages depending upon the power voltage requirements of the various monitored circuits **170**.

[0019] During a default state, system **150** is in a sleep mode to reduce power consumption. The awakening of system **150** may be responsive to a schedule. Alternatively, system **150** may awaken to check whether a user is commanding it to perform an automated monitoring of the monitored circuit(s) **170**. In some embodiments, a controller **165** may trigger or command power management circuit **160** to awaken and begin producing its power supply voltage(s) such as the 12 V power supply voltage or the telematics power supply voltage. Alternatively, power management circuit **160** may continue to produce its power supply voltages even while system **150** is in the sleep mode. In such constantly-on embodiments, the application of the 12 V power supply voltage to the monitored circuit **170** may be gated by a switch S1 that is controlled by controller **165**. Controller **165** would thus close switch S1 to activate the monitored circuit **170** so that its status may be determined and a resulting message about this status being transmitted by the telematics transceiver **175**. But switch S1 would be unnecessary if instead the power management circuit **160** collapsed (discharged and did not power) its power supply voltages during the sleep mode and would only produce the monitored circuit's power supply voltage upon command from controller **165**. A similar switch S**2** on the telematics transceiver's power rail is also optional.

[0020] Since the power supply voltages for the telematic transceiver **175**, the power management circuit **160**, controller **165**, and monitored circuit **170** are all ultimately derived from the battery power supply voltage during the active mode of operation for system **150**, system **150** may advantageously monitor a series of monitored circuits **170** despite the absence of the auxiliary power from the tractor. The monitored circuit **170** may be a digital circuit such as a sensor. System **150** may then receive a status directly from the monitored circuit **170**. Alternatively, system **150** may include a monitoring circuit as will be explained further herein that determines a status of a corresponding lamp in the towed asset. With the status determined through the selective powering of the monitored circuit **170**, the telematics transceiver **175** may then transmit the status to a remote user. For example, if the telematics transceiver **175** is a cellular transceiver, the cellular transceiver may transmit the status to a corresponding base station and ultimately to the Internet (the "cloud") to the remote user.

[0021] The resulting automated monitoring is quite advantageous. For example, the system **150** may be used to check the status of each lamp in the trailer: e.g., first the brake lights, then the license plate light, and so on with the remaining lamps. In addition, various sensors or other digital circuits within the trailer may be powered (simultaneously if they share the same power rail) and monitored despite the absence of the auxiliary power. For example, a user may remotely monitor the status and data from an ABS module, a TPMS receiver, a cargo sensor, trailer lamps, a door sensor, and so on. In this fashion, system **150** may be used to fully or partially verify the operating status of all the lamps, sensors, modules, etc. within the trailer despite the trailer being disconnected from a tractor and without auxiliary power. A user may be assured of an operating status of the trailer before sending a tractor to connect to the trailer.

[0022] The following discussion will be directed to the adaptation of a nose box to include the automated asset monitoring system **150**. However, it will be appreciated that the automated asset monitoring system **150** may be distributed or concentrated in other locations within the towable asset. The following discussion will thus be directed to a nose box implementation without loss of generality. The nose box integrates and protects the automated asset monitoring system **150**. The following description will be directed to nose boxes compatible with the Phillips i-Box™ form factor or the Phillips S7™ form factor. However, it will be appreciated that any suitable nose box form factor may benefit from the principles and techniques disclosed herein.

[0023] An example nose box **100** that is compatible with the Phillips i-Box™ form factor is shown in FIG. **1**B. Due to the inclusion of the automated asset monitoring system **150** discussed with regard to FIG. **1**A, nose box **100** may also be denoted as a "smart" nose box **100**. A housing **105** includes a J-560 coupler such as a Phillips Quick-Change socket QCS2® 125 that when opened may receive a 7-way coupler from the tractor. As implied by the term "nose," smart nose box **100** is placed at the front of a trailer. Housing **105** includes a lid **110** that may be rotated away from housing **105** on an axis **130** to expose an inner surface of lid **110** and an interior of housing **105**. Lid **110** may also include a solar panel **120** on its outer surface for recharging of a battery in smart nose box **100** as will be discussed further herein. In other embodiments, the solar panel may be remote from nose box **100**. Battery **156** may thus be a rechargeable battery as recharged through the solar panel power while the auxiliary power is not available. When the towable asset is connected to the tractor, battery **156** may be recharged using the auxiliary power. Housing **105** includes a removable grommet **135** so that a 7-way harness and other leads may exit housing **105** and extend to their trailer destinations.

[0024] Smart nose box **100** is configured to power and monitor various monitored circuits **170** in the towable asset such as the ABS module, a tire pressure monitoring system (TPMS) receiver, a tire inflation system, a cargo sensor, a door sensor, and so on despite the absence of the auxiliary power. Information regarding the status or signals from these monitored circuits may be transmitted to a user by a telematics transceiver **175** (FIG. **1**A) in smart nose box **100** as will be explained further herein. The status for a monitored circuit **170** may include whether the monitored circuit is operative, inoperative, open circuited, short circuited and so on. In addition, the status may include a message from a sensor such as whether the towable asset includes cargo, its weight, the tire pressure, the towable asset's location, and so on.

[0025] Nose box **100** is shown in FIG. **2** with the lid **110** opened away from housing **105** to expose a platform **205** in the interior of housing **105**. Platform **205** seals and protects a housing circuit board as will be discussed further herein. Platform **205** is also a convenient location for a plurality of terminals **210**. The following discussion will assume that terminals **210** are threaded posts so as to ease the attachment of a corresponding lead or wire by threadably engaging the posts with a corresponding nut but it will be appreciated that other types of terminals such as those requiring solder or a press fit may be used in alternative implementations. A 7-way wiring harness **215** is shown that connects between corresponding pins or terminals in the J-560 coupler **125** and corresponding posts **210** on platform **205**. In some embodiments, there are three rows of posts **210**. A top row of posts **210** receives the red, black, brown, green, and yellow wires from 7-way wiring harness **215**. Each wire connects to a corresponding post **210**. As known in the 7-way wiring arts, red is for the brake/stop lights, black is for the license plate lighting, brown is for the marker lights, green is for the left turn signal light, and yellow is for the right turn signal light. The corresponding lights in the trailer may also be denoted as lamps.

[0026] The top row of posts **210** receives the red, black, brown, green, and yellow wires from 7-way wiring harness **215**. The red, black, brown, green, and yellow wires from a trailer 7-way wiring harness (not illustrated) connect to the bottom row of posts **210**. The top row of posts **210** may thus be denoted as the upstream (tractor) posts whereas the bottom row of posts **210** may be denoted as the downstream (trailer) posts. This separation between upstream and downstream posts on the platform **205** is quite advantageous with regard to the monitoring of the corresponding lamps as will be further discussed herein.

[0027] The 7-way wiring harness **215** also includes a ground wire (which is typically colored white) that connects to a ground post **220**. Platform **205** may include two ground posts **220** that are shorted to each other through a ground strap or plate **225**. In addition, 7-way wiring harness **215** includes a blue wire (the auxiliary power lead) that couples to an auxiliary post **230** on platform **205** for powering an auxiliary circuit within the trailer. Because the tractor is not connected while system **150** is active, auxiliary post **230** does not have auxiliary power while system **150** functions in nose box **100** to monitor its monitored circuits **170**. As discussed for posts **210**, each of posts **220** and **230** may be threaded so that a wire fitting may be connected to the post by a nut or fastener that is screwed down on the threads. The blue wire in the 7-way wiring harness **215** as well as the corresponding blue wire in the trailer's 7-way wiring harness both share auxiliary post **230**. This is advantageous with regard to an accurate monitoring of the brake lights as will be explained further herein.

[0028] A housing circuit board **305** is affixed to a back side of platform **205** as shown in the exploded view of FIG. **3**A. Housing circuit board **305** includes a circular aperture **310** for each of posts **230**, **210**, and **220**. Each circular aperture **310** is surrounded by a conductive ring **315** on housing circuit board **305**. Platform **205** and housing circuit board **305** are shown in FIG. **3**A from the perspective of the tractor. This side of the housing circuit board **305** may be denoted as the front side whereas a back side faces the trailer. A partially cut-away exploded view of the back side of housing circuit board **305** and platform **205** from the perspective of the trailer is shown in FIG. **3**B. It may thus be seen that the posts extend through platform **205** to project from the backside of platform **205** and include threaded bores **325**. Each post thus includes a back-side portion that includes a threaded bore **325**. The back-side portion of each post is electrically connected to a front-side portion that extends above platform **205**. A screw **320** for each post extends from the backside of housing circuit board **305** through an aperture **310** to be received in the post's threaded bore **325**. As also seen in FIG. **3**B, the backside of housing circuit board **305** may also include a conductive ring **315** surrounding each aperture **310**. The resulting fastening of circuit board **305** to posts **210**, **230**, and **220** (FIG. **3**A) is quite advantageous in that no soldering is necessary to electrically couple circuits on circuit board **305** to corresponding posts. Despite the lack of soldering, circuitry on both the front and back sides of housing circuit board **305** can electrically couple to each post through a circuit board contact to the corresponding conductive ring **315**. Note that conductive rings **315** may be placed on just one circuit board surface around an aperture **310** in alternative embodiments as vias may be used that conduct from one side of the housing circuit board surface to another to allow circuitry on both sides of the housing circuit board to electrically couple to a conductive ring **315** and thus to the corresponding post. Note that the post platform **205** is configured to seal housing circuit board **305** within housing **105** to protect housing circuit board **305** from the elements. For example, a rim of post platform **205** may receive a gasket (not illustrated) to assist in the sealing of the rim of post platform **205** against a base of housing **105**. Housing **105** may be denoted as a "tub" because of its shape. The open bore of the tub is then enclosed by lid **110**.

[0029] The electrical coupling to each post is advantageous with regard to monitoring the lamps using corresponding monitoring circuits on housing circuit board **305**. For example, consider a monitoring (and activation) circuit **400** as shown in FIG. **4** that couples between a tractor-side conductive ring **315** (corresponding to one of the posts **210** in the top row of the posts in FIG. **2**) to a trailer-side conductive ring **315** (corresponding to one of the posts **210** in the bottom row of the posts). Each monitoring circuit **400** would be integrated onto either the front side and/or the back side of housing circuit board **305**. In monitoring circuit **400**, each of the corresponding posts **210** would correspond to the same lamp. There would thus be a monitoring circuit **400** between the tractor-side and trailer-side red wire posts **210** for the brake lights, another for the marker lights between the tractor-side and trailer-side brown wire posts **210**, and so on for the remaining lamps. Each monitoring circuit **400** includes a switch such as an NMOS transistor **M1** that is upstream of a current sense circuit **405** and a voltage monitor **415**. During normal operation (auxiliary power being present), a controller such as an micro-controller (MCU) **420** controls transistor **M1** to be conductive so that the tractor may control whether the corresponding lamp is illuminated or not. But during operation of system **150**, the auxiliary power is not present. When system **150** is active (and is monitoring the corresponding lamp), power management circuit **160** converts the battery voltage into the 12 V power supply voltage for the lamp being activated by monitoring circuit **400**. To prevent current from traveling upstream at that time, controller **420** may maintain transistor **M1** in an off state while system **150** monitors the lamp through monitoring circuit **400**.

[0030] With the lamp powered on from the power supply voltage from power management circuit **160**, a current sense circuit **405** measures the current supplied to the lamp since this current must pass through monitoring circuit **400** to drive the conductive ring **315** that ultimately couples to the lamp. Current sense circuit **405** may be a Hall sensor or may be a sense resistor. In addition, monitoring circuit **400** includes a fuse such as an e-fuse **410** to protect the trailer lamp from excessive currents.

[0031] In some embodiments, power management circuit **160** may couple to the lead between transistor **M1** and the current sense circuit **405** through the switch **S1** discussed with regard to FIG. **1**A. During the automated monitoring of a lamp by system **150** in such embodiments, the power management circuit **160** provides the lamp power signal through the closed switch **S1**.

[0032] Another controller such as microcontroller (MCU) **420** on housing circuit board **305** determines the health of the lamp being driven by monitoring circuit **400** by monitoring the sensed voltage and the sensed current. Should the MCU **420** detect that the voltage is too low, a problem is detected with regard to the power supply voltage coming from the power management circuit **160**. On the other hand, if the voltage is normal (typically 12 V) but the current is too low (or too high), either the corresponding lamp is malfunctioning or is broken. MCU **420** may then report the health of the corresponding lamp to a user using the telematics transceiver **175**, which may be located in lid **110** as will be explained further herein.

[0033] Referring again to FIG. **2**, note that the auxiliary post **230** is used to power an auxiliary circuit in the trailer such as an anti-lock-brake (ABS) module while the tractor is connected. The current draw of such a module is variable depending upon the mode of the ABS module. Housing circuit board **305** thus does not need a monitoring circuit **400** for auxiliary post **230** such that there is no need for a trailer-side auxiliary post **230** and a tractor-side auxiliary post **230**. But housing circuit board **305** may include a voltage monitor to verify whether the auxiliary post **230** is receiving the proper voltage from the tractor (e.g., 12 V) while the trailer is connected to the tractor.

[0034] To provide extra robustness for powering an ABS module, it is known to power the ABS module through both the trailer's blue wire and its red wire. In this fashion, if the auxiliary power from the tractor is absent such as due to a blown fuse, the ABS module will still be powered every time the brakes are applied from the resulting powering of the trailer's red wire. To accommodate this practice, the trailer-side red post in the top row of posts as shown in FIG. **2** also couples to a wire **250** that ends in a coupler **260**. A tractor-side wire (not illustrated) couples to coupler **260** and extends through grommet **135** to provide back-up power to the ABS module during normal operation (the tractor being connected to provide auxiliary power). Note that wire **250** does not couple to the trailer-side red post **210** because such a coupling would corrupt the current measurement by the monitoring circuit **400** for the brake lights should the ABS module draw power from the trailer's red wire instead of from the trailer's blue wire.

[0035] The voltage of auxiliary post **230** and the voltage/current health of the lamps as monitored by monitoring circuits **400** may be reported by MCU **420** to drive a data cable **140** that couples to a corresponding data cable **145** that is received by a telematics unit in lid **110**. Various other data signals may also be propagated through data cables **140** and **145**. A power and ground cable **159** couples from housing circuit board **305** to a coupler to drive a corresponding power and ground cable **155** for the telematics unit in lid **110**. The telematics unit or circuitry within lid **110** is powered by battery **156** as converted through power management circuit **160** while system **150** is active.

[0036] A trailer-side exploded view of an inner cover **505** for lid **110** is shown in FIG. **5**. To allow the reporting of telematics and the operation of system **150** while the trailer is disconnected from a tractor, inner cover **510** supports the rechargeable battery **156** (e.g., a lithium-ion battery). A lid circuit board **520** is attached to inner cover **510** and includes the telematics transceiver **175**. Any suitable wireless technology may be used for telematics transceiver **175** including cellular or WiFi. A cellular telematics transceiver **175** is particularly convenient due to the omnipresence of cellular coverage at virtually all warehouse locations and also due to the range of a cellular connection. The following discussion will thus assume that the telematics transceiver **175** is a cellular transceiver.

[0037] To conserve power, the default mode of the telematics transceiver **175** in the absence of the auxiliary power is a sleep mode in which the telematics transceiver **175** is powered down. Telematics transceiver **175** may be scheduled to periodically wake-up and check for commands to activate system **150** due to the receipt of a user command such as delivered over the internet (the "cloud") and then through a cellular connection to telematics transceiver **175**. In receipt of this command, the system **150** may perform its automated asset monitoring. In addition, the telematics transceiver **175** may be scheduled to periodically wake up to selectively power a monitored circuit **170** to receive data or determine its status and transmit the data and/or status through the cellular link and through the cloud to a user. One form of data to report may be the position of the trailer such as determined through a GPS receiver **535** that is also integrated onto lid circuit board **520**. In addition, lid circuit board **520** may include a local networking transceiver such as a Bluetooth transceiver **530** to receive data from Bluetooth-enabled sensors on the trailer. The trailer may include additional sensors and circuits that communicate with the controller (e.g., a MCU) **165** on lid circuit board **520** when activated by system **150**.

[0038] With regard to this communication, platform **205** includes a center row of posts **270**. A pair of posts **270** connect through housing circuit board **305** and data cables **140** and **145** to MCU **165**. MCU **165** is configured to support a Control Area Network (CAN) bus transceiver for this pair of posts **270**. A CAN bus routed from the trailer through grommet **135** may then be wired to this pair of posts **270** to couple to CAN-bus-enabled sensors in the trailer. Another pair of posts **270** may correspond to a RS 485 bus that is also routed to appropriate sensors in the trailer. Regardless of whether a sensor is connected to the RS 485 bus or to the CAN bus, the sensor may receive power from another one of posts **270** that functions as a sensor power post during the automated monitoring. For example, referring again to FIG. **1**A, the activation of power management circuit **160** allows a 12 V power supply to energize the sensor power post and thus power the various monitored sensors. The CAN bus and the RS 485 bus is also routed through the corresponding posts **270** and through data cables **140** and **145** to MCU **165**. In this fashion, a wide variety of sensors in the trailer may be networked to MCU **165** during the operation of system **150**. For example, the TPMS receiver, a cargo sensor, a door sensor, a weight sensor, and so on may be powered and networked to MCU **165**. Other sensors may be networked to MCU **165** through a Bluetooth connection with Bluetooth transceiver **530**.

[0039] In addition to the coupling through the CAN and proprietary buses for receiving data and sensor status, the trailer's auxiliary lead may be used to transmit data and commands using a power line communications (PLC) module **515** during normal operation (the tractor being connected and providing auxiliary power). For example, the ABS module may report error conditions over the auxiliary lead to auxiliary post **230**. Circuit board **305** receives auxiliary post **230** in a corresponding conductive ring **315** so that the auxiliary post PLC signaling may be coupled over data cables **140** and **145** to PLC module **515**. PLC module **515** decodes the PLC signaling to provide the data to MCU **165** so that it may then be transmitted by the telematics transceiver **525** to a user in the cloud. Conversely, MCU **165** may transmit a query to PLC module **515** that is then translated to the PLC protocol and transmitted through data cables **140** and **145**, circuit board **305**, auxiliary post **230** and its corresponding ring **315** to the auxiliary lead in the trailer. The query may be a command to the ABS module to provide a status.

[0040] Regardless of the type of buses and whether PLC module **515** is included, note the advantages of the battery **156**. Even with the trailer disconnected from the tractor's 7-way coupler or connected but with the auxiliary lead (the blue wire) in the tractor's 7-way coupler without power, telematics transceiver **525** may periodically wake up for an activity period to either listen for commands from a user such as downloaded from the cloud over a cellular link to the telematics transceiver **525** or to periodically wake up and provide a GPS location.

[0041] Lid circuit board **520** also contains the power management circuit **160**. When auxiliary post **230** has power, power management circuit **160** may be configured to use auxiliary post power from auxiliary post **230** as routed through the corresponding conductive ring **315**, housing circuit board **305**, and power cables **159** and **155** to power the remaining electronics on. Since the auxiliary post voltage is 12 V, power management circuit **160** may include a DC-DC switching power converter such as a buck converter to convert the battery voltage into a lower, more suitable DC power supply voltage (e.g., a telematics power supply voltage) for the lid electronics.

[0042] Because the telematics circuitry may be integrated into lid **110**, a user may retrofit a conventional nose box with lid **110** should the user not desire the monitoring of the lamps as discussed with regard to circuit board **305** in housing **105**. The sensors will still be powered by the generated 12V power supply voltage during such an automated asset monitoring mode of operation. A conventional nose box housing may merely have the posts for a 7-way cabling to the trailer lamps and auxiliary circuit. Since there would be no sensor power post in such a conventional nose box house, power cable **155** may couple to an adapter to a trailer power lead to provide the generated 12 V power supply voltage during the automated asset monitoring mode of operation.

[0043] As those of some skill in this art will by now appreciate and depending on the particular application at hand, many modifications, substitutions and variations can be made in and to the materials, apparatus, configurations and methods of use of the devices of the present disclosure without departing from the scope thereof. In light of this, the scope of the present disclosure should not be limited to that of the particular embodiments illustrated and described herein, as they are merely by way of some examples thereof, but rather, should be fully commensurate with that of the claims appended hereafter and their functional equivalents.

We claim:

1. An automated towable asset monitoring system having an active mode of operation in which a towable asset is not coupled through a 7-way coupler to a tractor and an inactive mode of operation in which the towable asset is coupled through the 7-way coupler to a tractor, the automated asset monitoring system comprising:

a battery;

a power management circuit configured to convert a battery voltage from the battery to a monitored circuit power supply voltage during the active mode of operation and to collapse the monitored circuit power supply voltage during a sleep mode while the towable asset does not receive an auxiliary power from the 7-way coupler;

a controller configured to command the power management circuit to power a monitored circuit in the towable asset during the active mode of operation with the monitored circuit power supply voltage and to prevent the monitored circuit power supply voltage to power the monitored circuit during the inactive mode of operation;

a telematics transceiver configured to transmit data regarding an operation of the monitored circuit during the active mode of operation to a user remote from the towable asset, wherein the system is integrated into a nose box that includes:

a housing;

a terminal platform within the housing, the terminal platform including a first plurality of terminals for receiving lamp power signals from the 7-way coupler, a second plurality of terminals for transmitting the lamp power signals to lamps in the towable asset; and

a housing circuit board associated with the terminal platform, the housing circuit board including a plurality of lamp monitoring circuits, wherein the controller is further configured to force the power management circuit to supply an active one of the lamp monitoring circuits with the monitored circuit power supply voltage during the active mode of operation to power a respective one of the lamps in the towable asset through a respective terminal in the second plurality of terminals, the active one of the monitoring circuits being configured to measure a current and a voltage supplied to the respective one of the lamps to determine an operating condition of the respective one of the lamps, and wherein the data includes the operating condition.

2. The system of claim **1**, further comprising:

a lid for enclosing an interior of the housing, the lid including an inner cover configured to enclose the battery and a lid circuit board including the telematics transceiver, the controller, and the power management circuit.

3. The system of claim **1**, further comprising:

a solar panel, wherein the battery is a rechargeable battery configured to be recharged from power delivered by the solar panel during the active mode of operation.

4. The system of claim **1**, further comprising:

a sensor power terminal, wherein the controller is further configured to command the power management circuit to provide the monitored circuit power supply voltage to the sensor power terminal during the active mode of operation to power an at least one sensor in the towable asset.

5. The system of claim **4**, further comprising a plurality of control area network (CAN) bus terminals, and wherein the controller is configured to receive a sensor signal through the plurality of CAN bus terminals from the at least one sensor, and wherein the data includes the sensor signal.

6. The system of claim **1**, wherein the monitored circuit comprises at least one of a lamp, a cargo sensor, a tire pressure inflation system sensor, a trailer weight sensor, or a camera.

7. The system of claim **1**, wherein the terminal platform is configured to seal the housing circuit board within the housing.

8. The system of claim **1**, wherein the first plurality of terminals comprises a first plurality of posts and the second plurality of terminals comprises a second plurality of posts.

9. The system of claim **2**, wherein the lid circuit board further includes a GPS receiver configured to be powered from the battery during the active mode of operation, and wherein the controller is further configured to control the telematics transceiver to periodically transmit a location from the GPS receiver during the active mode of operation.

10. A system for monitoring a lamp in a towable asset, comprising:

a rechargeable battery;

a power management circuit configured to charge the rechargeable battery using auxiliary power while the towable asset is coupled to a tractor, the power management circuit being further configured to generate a lamp power supply voltage during an active mode of operation in which the towable asset does not receive the auxiliary power;

a monitoring circuit configured to measure the lamp power supply voltage and a current delivered to the lamp to determine an operating condition of the lamp;

a controller configured to command the power management circuit to supply the lamp power supply voltage to the monitoring circuit during the active mode of operation,

a telematics transceiver configured to transmit a message regarding the operating condition to a user remote from the towable asset;

a first terminal for receiving the auxiliary power while the towable asset is connected to the towable asset; and

a second terminal for driving a lamp power lead connected to the lamp, wherein the monitoring circuit comprises a current sensor coupled between the first terminal and the second terminal and also comprises a switch coupled between the current sensor and the first terminal, wherein the controller is further configured to switch off the switch during the active mode of operation.

11. A system for monitoring a lamp in a towable asset, comprising:

a rechargeable battery;

a power management circuit configured to charge the rechargeable battery using auxiliary power while the towable asset is coupled to a tractor, the power management circuit being further configured to generate a lamp power supply voltage during an active mode of operation in which the towable asset does not receive the auxiliary power and to collapse the lamp power supply voltage during a sleep mode while the towable asset does not receive the auxiliary power;

a monitoring circuit configured to measure the lamp power supply voltage and a current delivered to the lamp to determine an operating condition of the lamp;

a controller configured to command the power management circuit to supply the lamp power supply voltage to the monitoring circuit during the active mode of operation,

a telematics transceiver configured to transmit a message regarding the operating condition to a user remote from the towable asset;

a first terminal for receiving the auxiliary power while the towable asset is connected to the towable asset; and

a second terminal for driving a lamp power lead connected to the lamp, wherein the monitoring circuit comprises a current sensor coupled between the first terminal and the second terminal and also comprises a switch coupled between the current sensor and the first terminal, wherein the controller is further configured to switch off the switch during the active mode of operation.

12. The system of claim **11**, wherein the power management circuit is further configured to supply the lamp power supply voltage to a node in the monitoring circuit between the switch and the current sensor.

13. The system of claim **12**, wherein the controller is further configured to switch on the switch while the towable asset is connected to the auxiliary power.

14. The system of claim **12**, wherein the controller is further configured to switch on the switch while the towable asset is connected to the auxiliary power.

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